

$$\mathcal{E}_i = -\frac{d\varphi}{dt}; \quad d\varphi = B dS; \quad dS = dy - 2x$$

$$d\varphi = B dy - 2x; \quad a = \frac{d\varphi}{dt}$$

$$\varphi = \frac{2B}{\sqrt{k}} \int \sqrt{y} dy = \frac{2B}{\sqrt{k}} \cdot \frac{y^{\frac{3}{2}}}{\frac{3}{2}} + C = 0$$

$$\int dv = \int a dt;$$

$$v = at + C = 0$$

$$v = \frac{dy}{dt}; \quad dy = at dt$$

$$y = a \int t dt = \frac{at^2}{2}$$

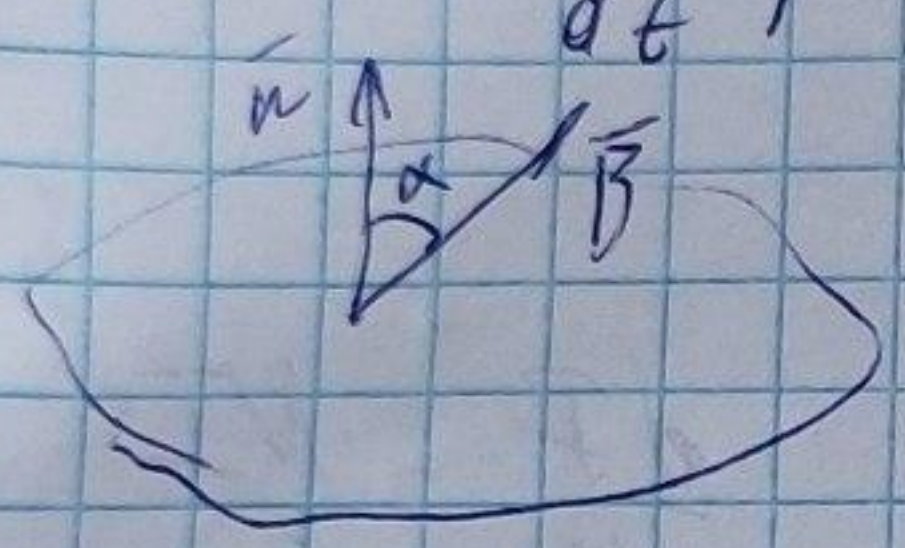
$$\varphi(t) = \frac{2B}{\sqrt{k}} \cdot \frac{2(at^2)^{\frac{3}{2}}}{2^{\frac{3}{2}} \cdot 3} = \frac{\sqrt{2} B + a^{\frac{3}{2}}}{\sqrt{k} \cdot 3}$$

5.3.3

$$\int_a^{a+b} d\varphi = \int_a^{a+b} \frac{A}{x} \frac{(bc+c-cx)dx}{b}$$

$$\begin{aligned} \varphi &= \frac{A}{x} \left(\int_a^{a+b} \frac{bc}{x} dx + \int_a^{a+b} \frac{cx}{x} dx - \int_a^{a+b} c dx \right) = \\ &= \frac{A}{x} \left(bc \ln x \Big|_a^{a+b} + a c \ln x \Big|_a^{a+b} + cx \Big|_a^{a+b} \right) \end{aligned}$$

$\mathcal{E} = - \frac{d\varphi}{dt}$, a, R, β ; $Q = ?$ (some work n.)

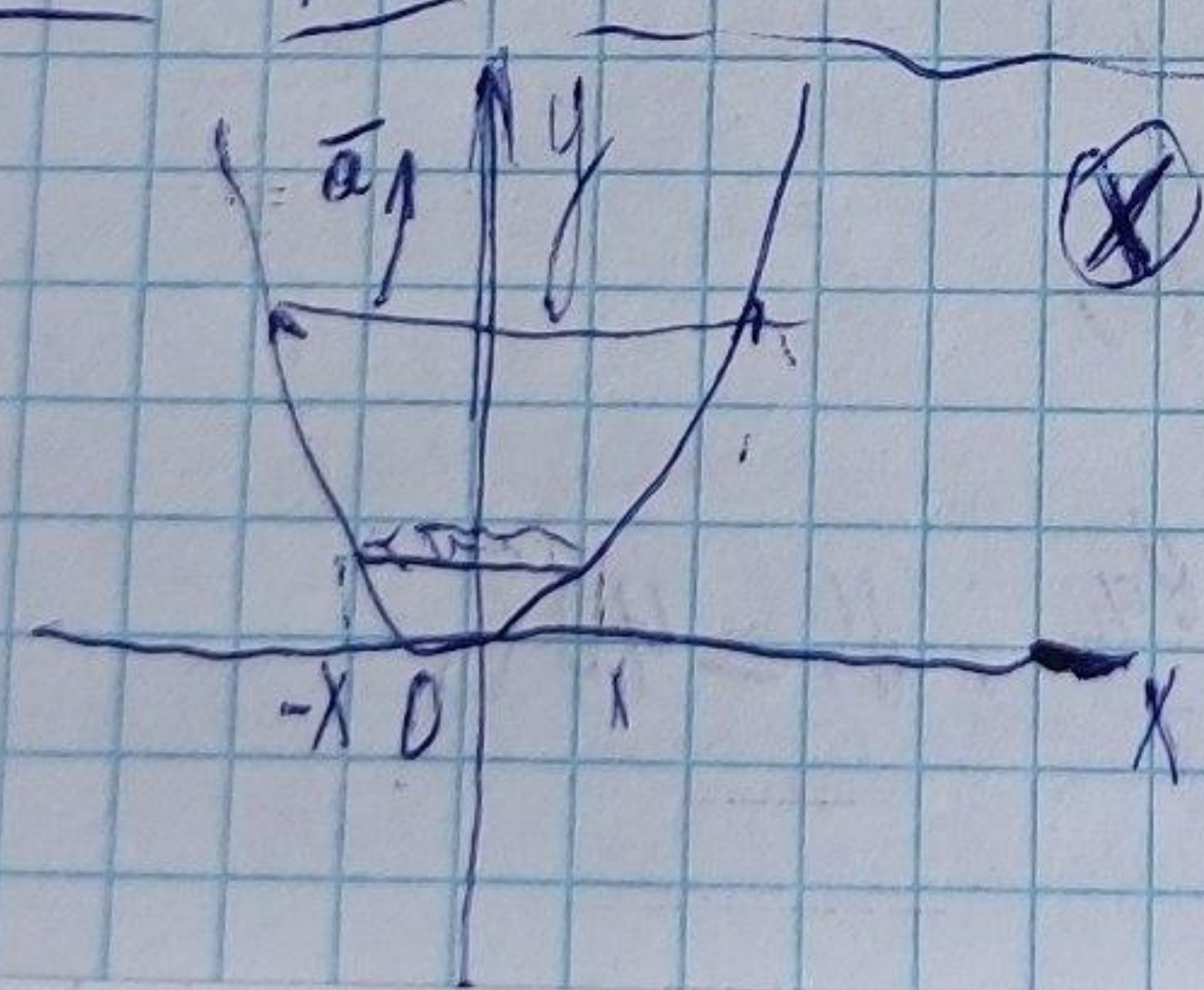


$$\begin{cases} \mathcal{E} = - \frac{d\varphi}{dt} \\ I = \frac{\mathcal{E}}{R} \\ I = \frac{d\varphi}{dt R} = \frac{dQ}{dt} \end{cases}$$

$$\varphi = \beta a^2 \cos x$$

$$dQ = I dt = \frac{\mathcal{E}}{R} dt = - \frac{d\varphi}{R}$$

$$\int_{\varphi_0}^0 dQ = \int_{\varphi_0}^0 \frac{1}{R} d\varphi = \frac{1}{R} \cdot (\beta a^2 \cos x - 0) = \frac{\beta a^2 \cos x}{R}$$



$y = kx^2$; $y(0) = 0$; a, β
 $\varphi(0) = 0$; $\mathcal{E}(y) = ?$